MUHAMMAD FAHAD MUSTAFA

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Education

Bachelor of Science in Computer Science (BS CS) – 4th Semester Student, 09/2024 – 06/2028
COMSATS University Islamabad, Sahiwal
Academic Standing: Cumulative Grade Point Average (CGPA): 3.84 / 4.00

ICS (Intermediate in Computer Science), *Punjab Group of Colleges (PGC)* 2022 – 2024

Matriculation in Sciences, *Cathedral Christ Church School Okara* 2020 – 2022

Technical Skills

Core Programming
Java (Advanced OOP), C++ (High-Performance Systems), Python (Automation & AI Stack)

Web & E-Commerce
HTML5, CSS3, WordPress (CMS Configuration), Shopify (Store Setup & Administration)

Algorithms & Databases
Data Structures & Algorithms (DSA), Graph Theory, Dijkstra's Path Optimization, Oracle DB (JDBC)

Mobile Development
Flutter Framework, Dart Language, Android App Architecture & Lifecycle Management

AI, RAG & Networks
Retrieval-Augmented Generation (RAG), Automated Pipelines, VLAN Configuration, Network Segmentation

Projects

AI-Driven Professional SaaS Product & Automation Pipeline, In Progress
Python, Advanced RAG Pipelines & Codebase Security
- Designing and building a professional-grade SaaS product codebase engineered via advanced automated development workflows.
- Implementing a comprehensive "God-Mode Automation" pipeline to orchestrate application backend logic, responsive frontend interfaces, and multi-tier data security wrappers.
- Developing high-accuracy contextual processing architectures using Retrieval-Augmented Generation (RAG) models to index and query production text corpuses.

Smart Office IoT Network Architecture, In Progress
Cisco Packet Tracer, Managed Firewalls & Enterprise Security
- Modeling and developing a highly secure corporate smart office IoT network infrastructure within a simulated network environment.
- Implementing network protection layers by deploying edge routers, managed firewalls, and strict macro-segmentation mapping via customized VLAN profiles.
- Debugging live network layout blocks, interface isolation mismatches, and packet routing anomalies to guarantee continuous enterprise uptime.

Pharmacy Management System, *Java, JDBC & Oracle Database RDBMS*

- Built a robust client application utilizing pure Java Object-Oriented Programming (OOP) methodologies including inheritance, polymorphism, and encapsulation for systematic data modularity.
- Developed a reliable data persistence layer by establishing secure network handshakes directly to an Oracle Database instance using JDBC API structures.

Hospital Shortest Path Routing Project, *C++ & Graph Analytics Engine*

- Modeled and built a spatial navigation simulator mapped within a custom directional graph layout using the native C++ compiler environment.
- Programmed an integration of Dijkstra's Shortest Path Algorithm to dynamically calculate minimum travel costs between clinical nodes with minimal execution latency.

School Management System, *C++ & DSA Core Architecture*

- System Implementation: Developed a lightweight administrative software system client in C++ to capture, structure, and organize student record repositories.
- Custom Data Structuring: Deployed custom Data Structures & Algorithms (DSA) logic maps to regulate running memory arrays and speed up data retrieval operations.